Lorenzo Remy Design Portfolio- CAD- Design for Manufacturing (DFM)

Overview

This portfolio showcases a range of design projects that emphasize Design for Manufacturing (DFM) principles and collaborative engineering processes. The projects span machined, sheet metal, and injection molded parts, as well as advanced CAD design for 3D-printed components used in high-powered rocketry. The CAD modelling showcased in this document was created in SolidWorks. Each section highlights the design intent, manufacturing considerations, and teamwork involved in achieving functional, manufacturable solutions.

Preface

This portfolio includes three parallel design studies for a computer assembly, each exploring a different manufacturing process: machining, sheet metal fabrication, and injection molding, to produce the same functional part. Due to limited customer-provided specifications, each design required strategic assumptions and process-specific optimizations.

The latter projects demonstrate collaborative, cross-functional engineering efforts, including integrated sheet metal-molded assemblies and cradle-to-grave 3D-printed aerospace components. These efforts reflect a strong emphasis on iterative design, stakeholder alignment, and practical application of DFM principles across diverse fabrication methods.

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Machined Part Design

Design Intent

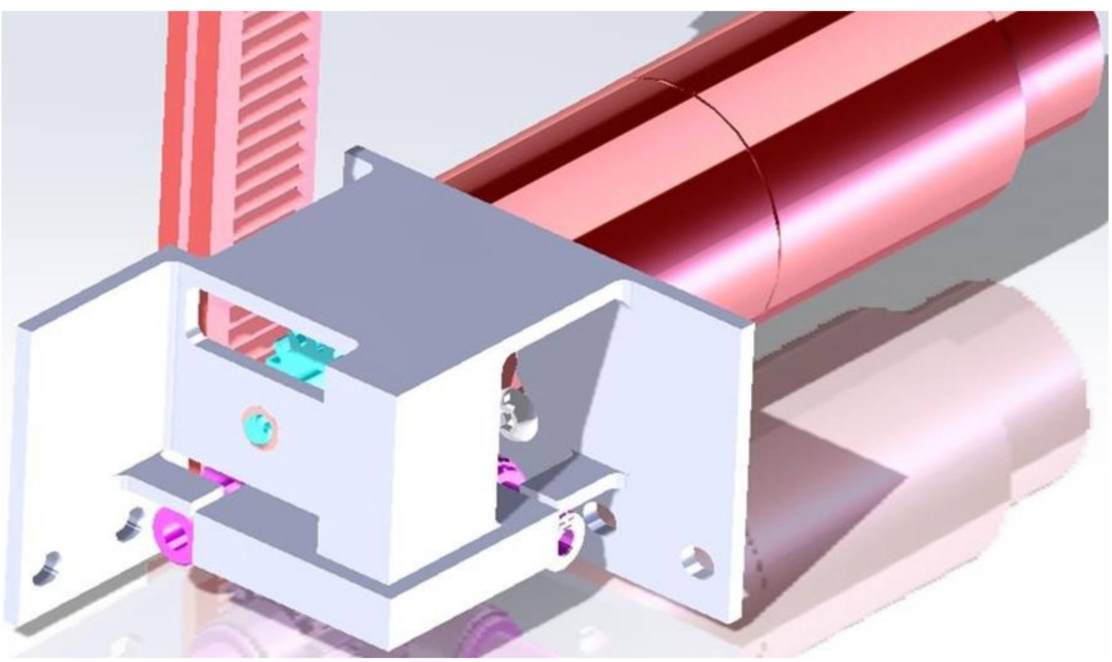
- Developed a machined aluminum part with features for precise assembly and structural rigidity.
- Included slots, cutouts, and lips for alignment, bearing support, and fastener access.
- Leveraged existing stock material to minimize waste and reduce the need for additional components.

Key Features

- **Slot and circle cuts:** Provide access to M3 screw holes and press-fit bearing support.
- **Cutouts and lips:** Allow for set screw access and precise part alignment.
- **Bearing holding plates:** Identical protrusions for secure bearing placement.
- **Rack alignment protrusion:** Ensures proper fit and fastening.

DFM Considerations

- Utilized five fixture positions for comprehensive milling, ensuring feature accuracy and minimizing part chatter.
- Sequenced milling operations to avoid over-cutting and preserve critical dimensions.
- Required to use Al 6061 for its machinability and mechanical properties.



Machined Part in Assembly

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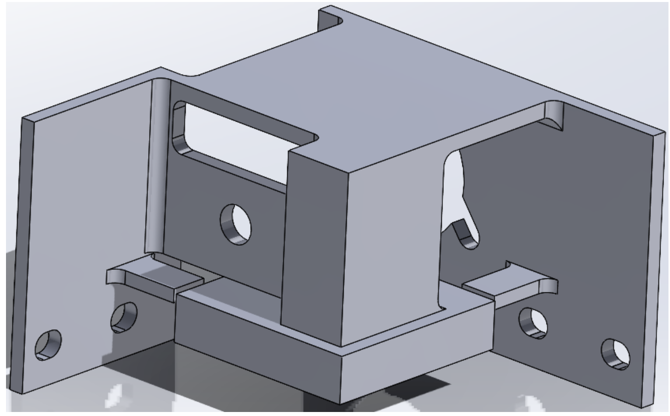
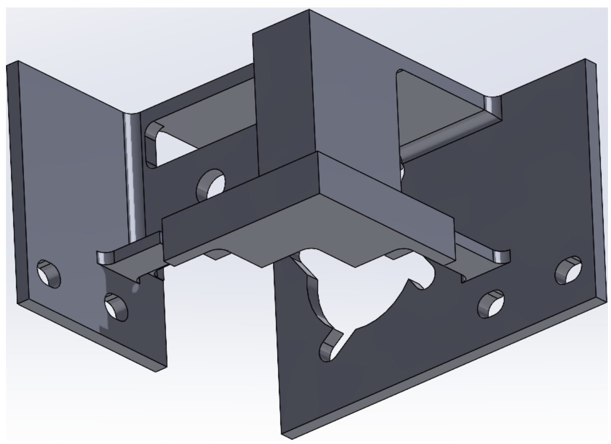
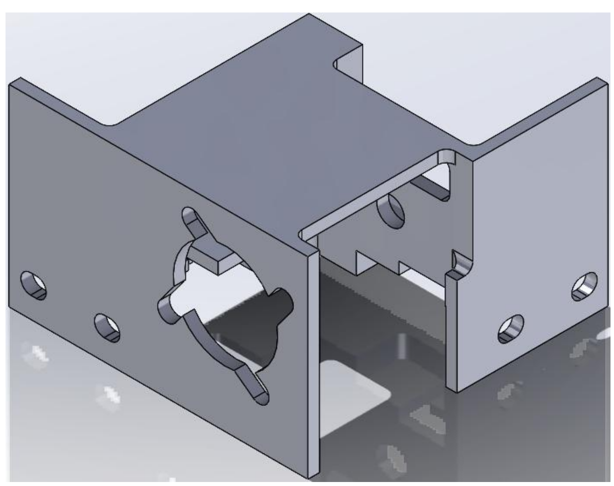


Machined Part from Front Top Right View Showing Features



Machined Part from Front Botton Right View Showing Features



Machined Part from Rear Top Right View Showing Features

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Sheet Metal Part Design

Design Intent

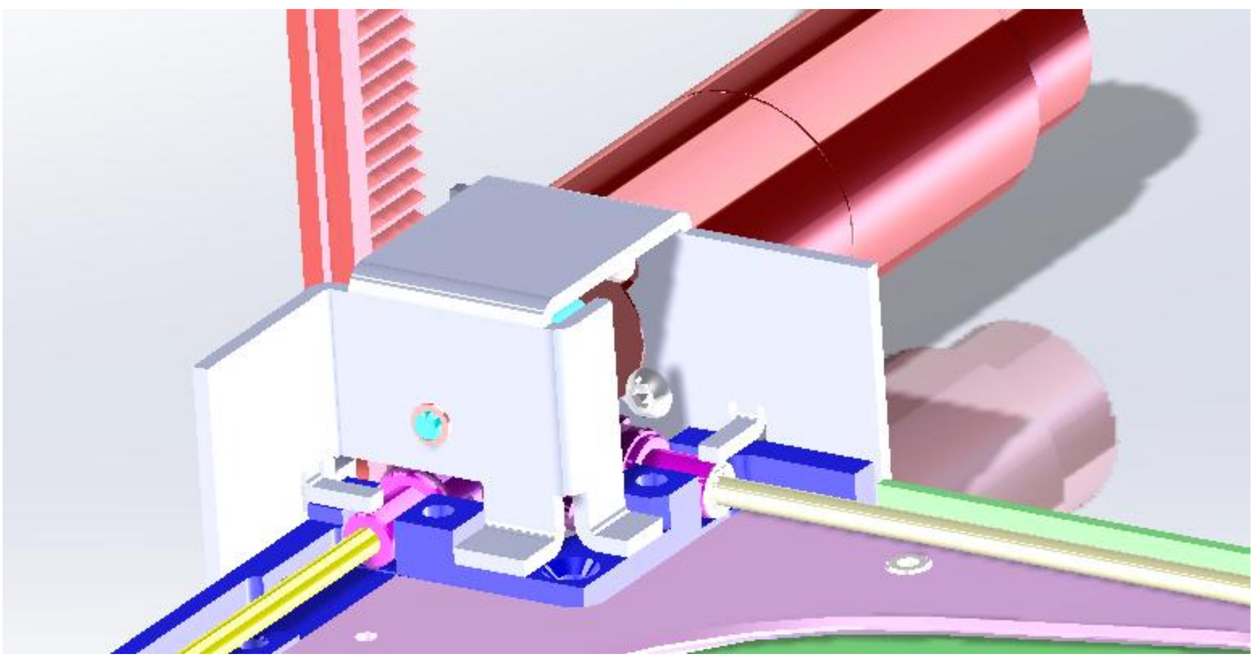
- Created a sheet metal component to serve as a mounting and alignment platform in an assembly.
- Incorporated bends, slots, and cutouts for fastener access, bearing placement, and part rigidity.

Key Features

- **Offset and protrusions:** Accommodate rack and align with assembly components.
- **Slots and holes:** Enable secure mounting and bearing press-fit.
- **Diving boards:** Firmly hold bearings in place.
- **Bends:** Add rigidity and facilitate assembly integration.

DFM Considerations

- Designed with 14-gauge Al 6061 sheet for optimal strength and manufacturability.
- Sequenced bends and stamping to ensure accurate final geometry.
- Allowed for variable mounting options to accommodate different installation environments.



Sheet Metal Part in Assembly

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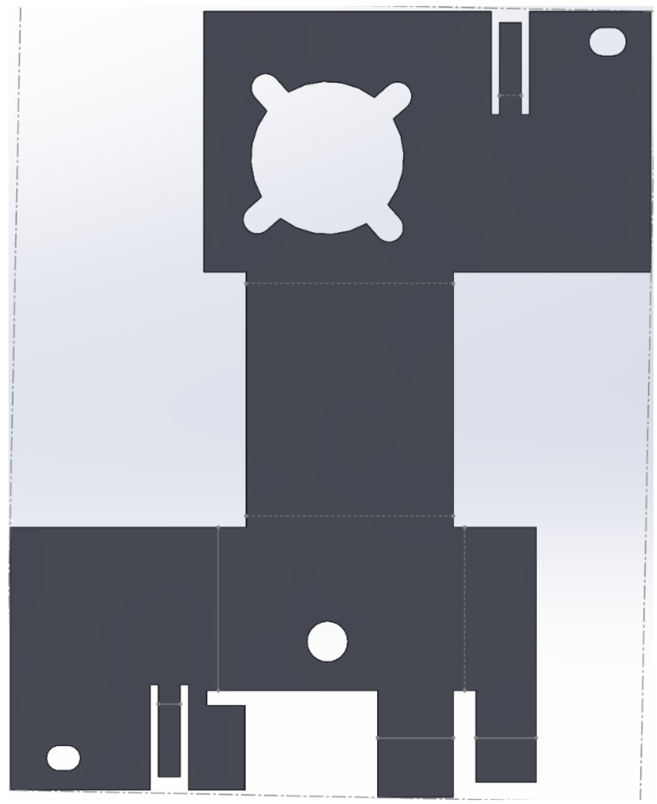
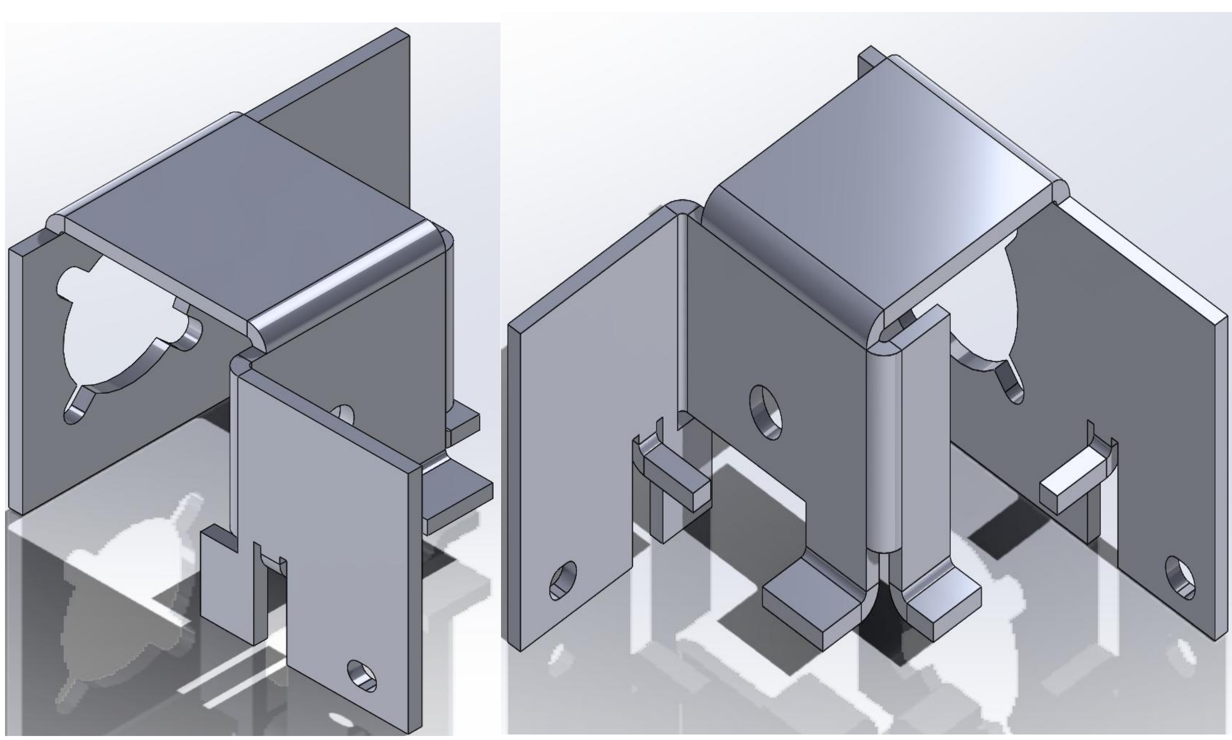


Flattened Sheet Metal Part



Folded Sheet Metal Part in Various Views Showing Features

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Injection Molded Part Design

Design Intent

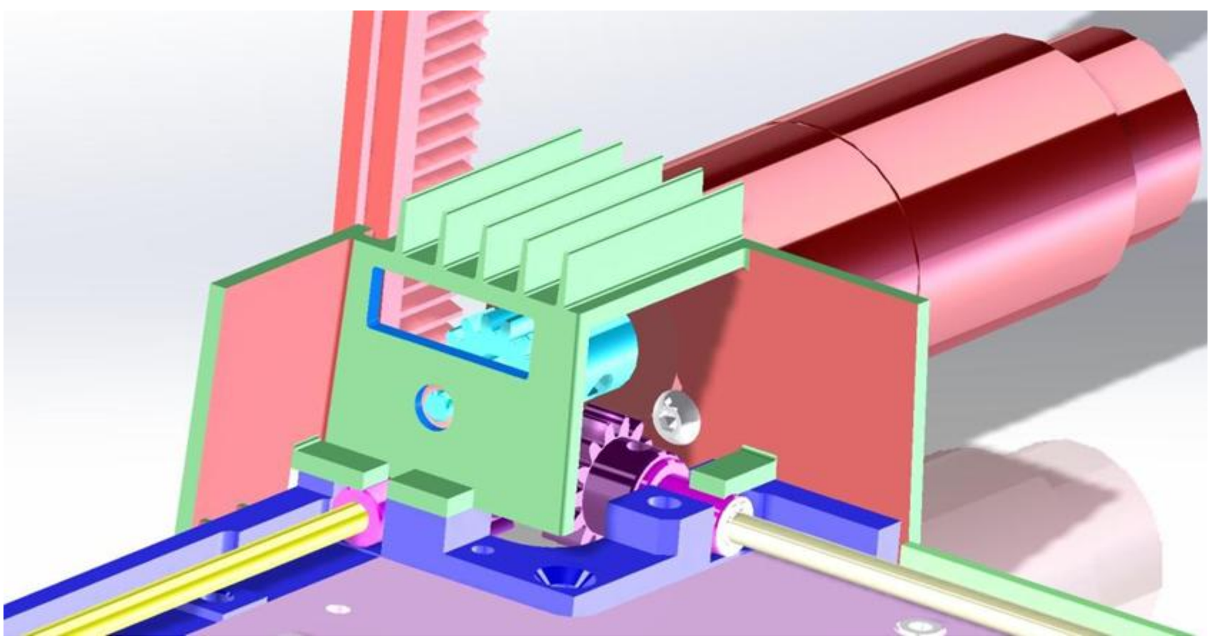
- Developed a part using Liquid Crystal Polymer (LCP) with uniform wall thickness and draft angles for efficient molding.
- Integrated ribs, alignment tabs, and relief cuts to enhance rigidity and assembly fit.

Key Features

- **Slots and ramps:** Provide access and offset drafts for flat contact surfaces.
- **Diving boards and ribs:** Secure bearings and add rigidity.
- **Alignment tabs:** Ensure precise assembly integration.

DFM Considerations

- Maintained 2mm wall thickness for even cooling and minimal warpage.
- Selected LCP for its high Young's modulus, enabling a 2 mm-thick injection-molded cantilever beam to meet the required flexure force (≥15 lbf and ≤35 lbf); alternative materials would require excessive thickness, hindering cooling efficiency, LCP achieved 17.8 lbf at target thickness.
- Applied 2° draft angles to facilitate part ejection and prevent undercuts.
- Utilized slide actions in the mold to create complex features without secondary operations.



Injection Molded Part in Assembly

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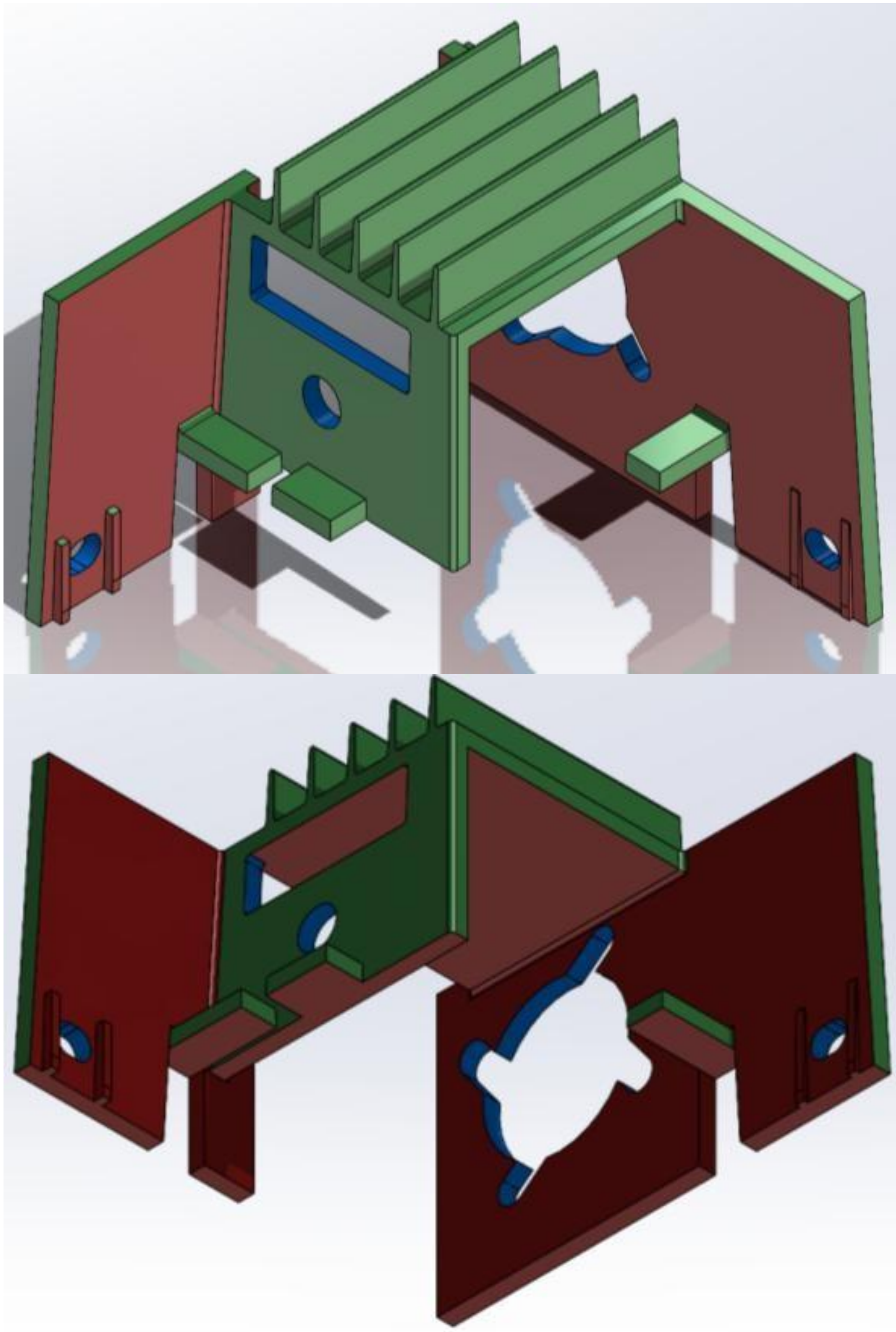

Injection Molded Part in Multiple Views Showing Cavity Mold (Green), Core Mold (Red) and Slide Action (Blue) Features

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Collaborative Design- Sheet Metal & Machined Part Integration

Design Intent

- Developed a **sheet metal mounting plate** designed to integrate seamlessly with both injection molded and machined components in a complex assembly.
- Prioritized features for **fastener access, mounting versatility, wire management, and safety enclosures** to ensure robust functionality and ease of assembly.

Key Features

- **Mounting holes and slots**: Enable secure installation and precise alignment with other components.
- **Openings for power and wire management**: Allow for external connections and organized cable routing.
- **Safety enclosures and teeth-like patterns**: Provide user safety and facilitate specific functions (e.g., easy tearing of paper towels).
- **Bearing and shaft mounts**: Ensure proper alignment and smooth operation of moving parts.

DFM Considerations

- **Material selection**: Used Al 6061 sheet for its balance of strength, lightness, and manufacturability.
- **Process sequencing**: Optimized the order of cutting, bending, and welding operations to maintain dimensional accuracy and minimize rework.
- **Uniform thickness**: Maintained consistent sheet thickness to simplify fabrication and avoid the need for additional welded elements.
- **Hardware integration**: Incorporated threaded inserts and robust connectors to compensate for the thinness of sheet metal, ensuring strong and reliable assembly.
- **Minimized part count**: Where possible, combined functions into single components to reduce assembly complexity and cost.

Collaborative Highlights

- **Cross-disciplinary coordination**: Worked closely with electrical, and injection molded manufacturing teams to finalize mounting patterns, enclosure features, and assembly steps.
- **Integrated feedback**: Adjusted design based on input from teams, ensuring manufacturability and ease of installation.

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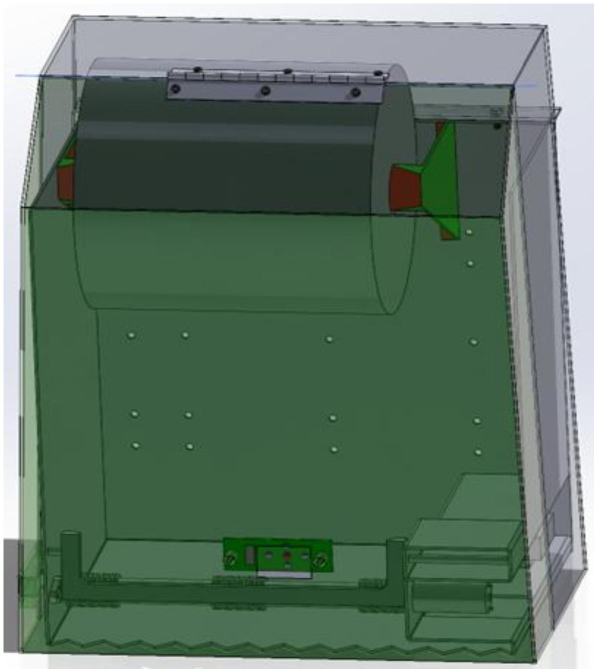
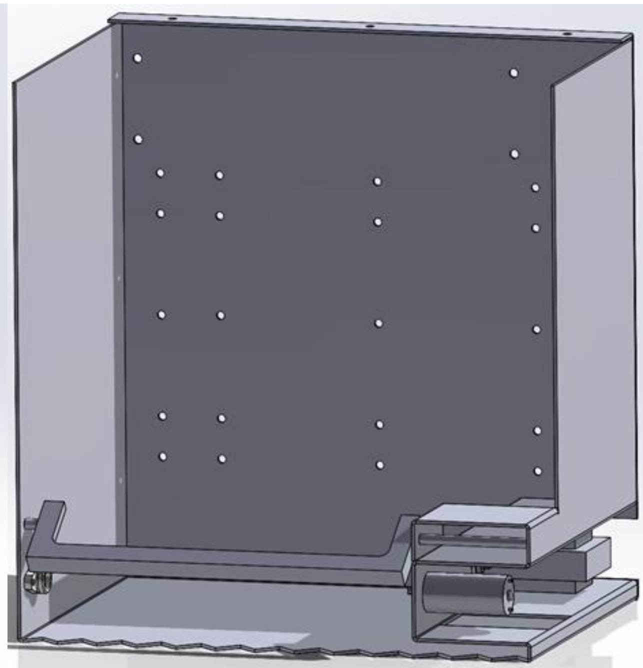
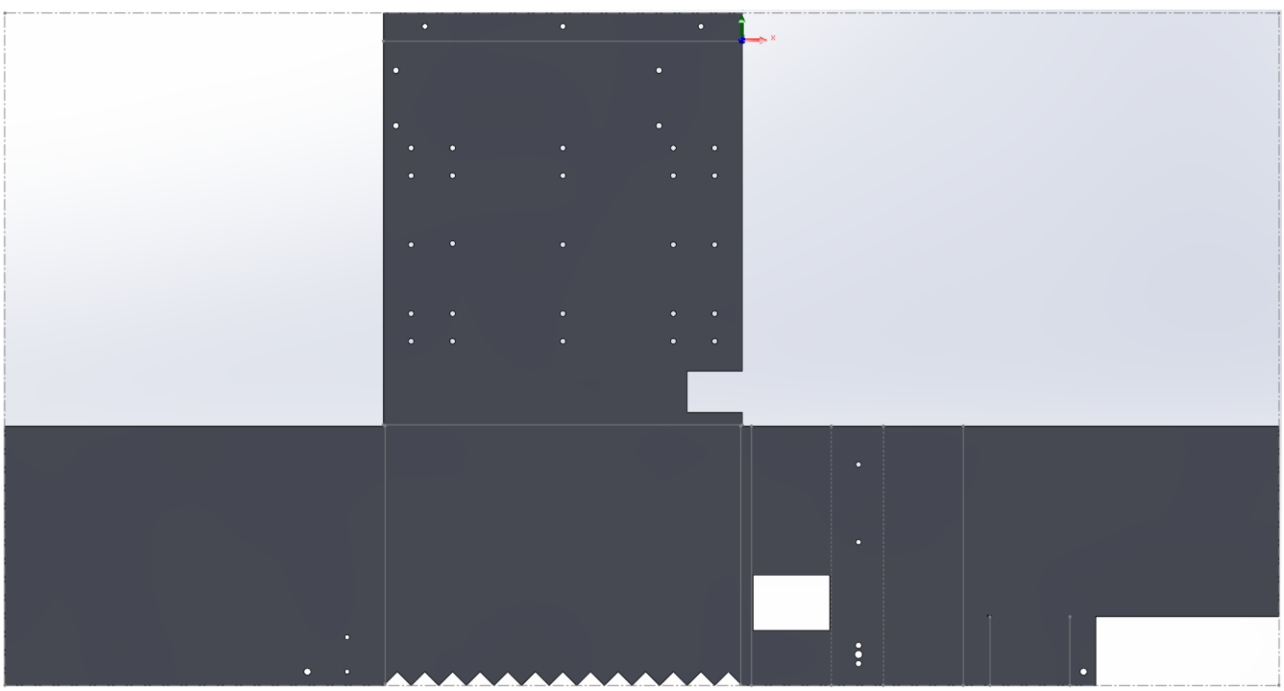
- **Documentation**: Provided clear assembly instructions and diagrams, highlighting critical alignment and safety steps.





Sheet Metal & Machined parts in Assembly



Flattened Sheet Metal Part

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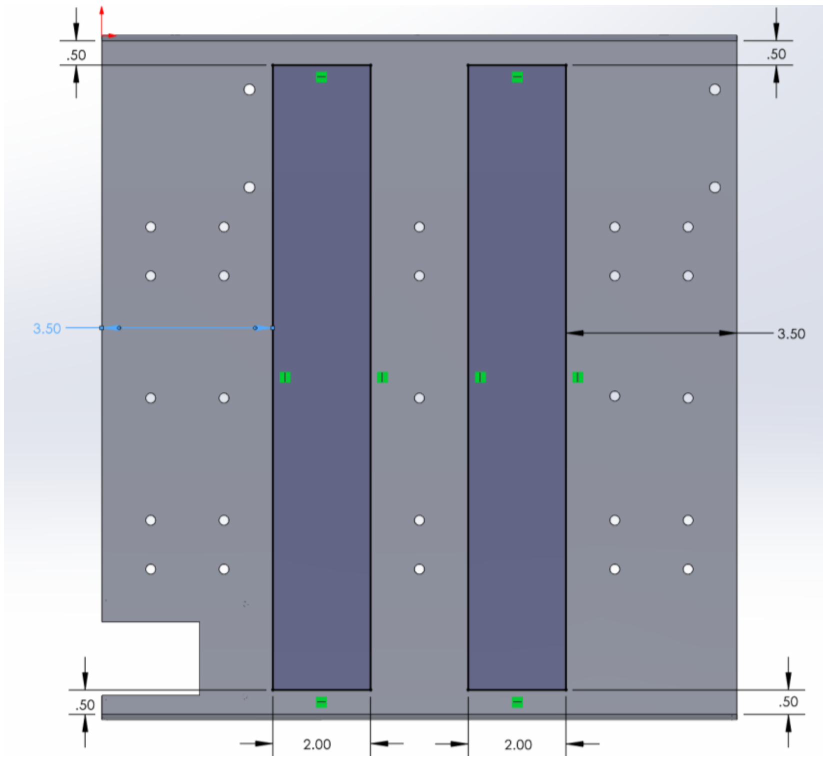
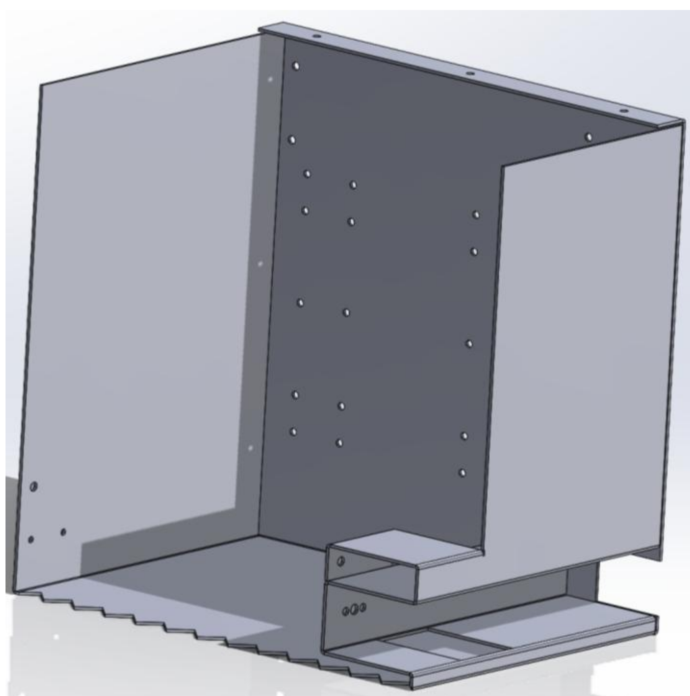


Rear of Sheet Metal Part Showing Mounting Surface



Folded Sheet Metal Part

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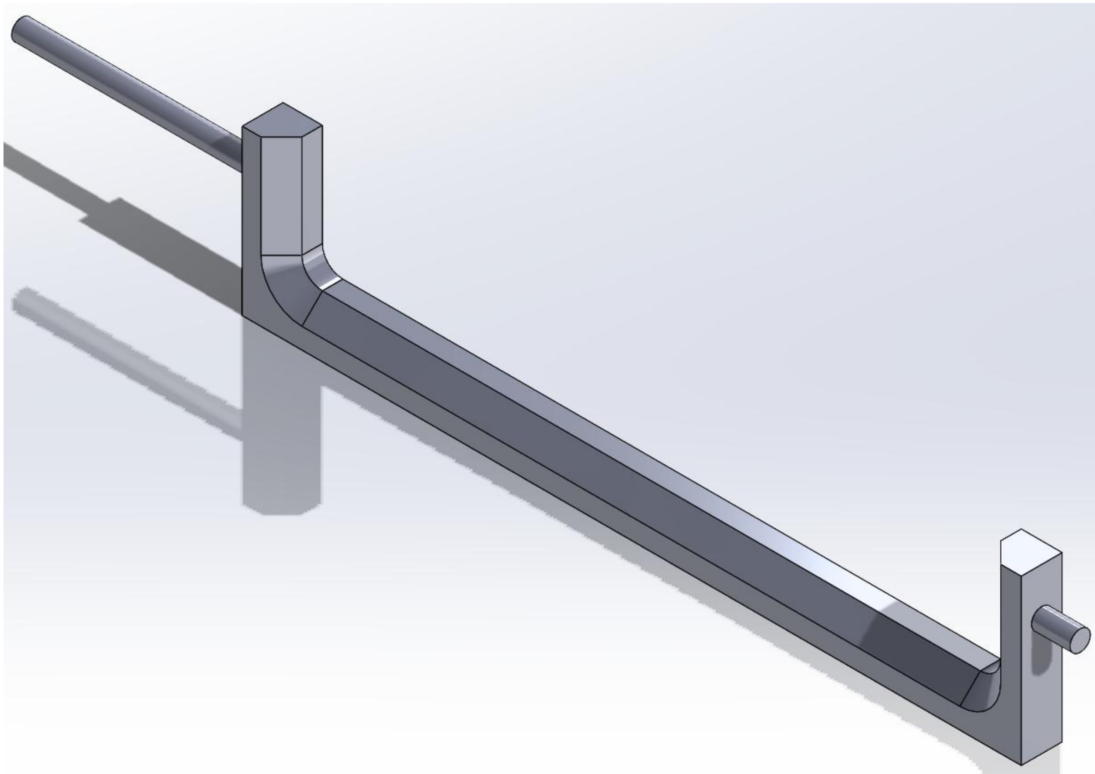

Machined Part in Isometric View Showing Features

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Collaborative Design: 3D Printed Parts- Cradle to Grave

Design Intent

- Developed custom PETG 3D-printed avionics tray and solid propellant engine casing for a high-powered solid propellant rocket.
- Focused on efficient payload packaging and maintaining the vehicle's center of gravity (CG) for flight stability.
- Created multiple CAD model iterations to refine fit, function, and manufacturability.

Key Features

- **Payload packaging:** Ensured efficient use of space and optimal CG for stability.
- **Iterative CAD models:** Demonstrated design evolution and response to feedback.
- **PETG material:** Selected for strength, flexibility, and printability under high-stress conditions. Additionally, PETG was selected over standard PLA due to its increased rigidity and superior performance in functional prototypes.

Avionics Tray/ Payload Packaging: Cradle-to-Grave Approach with Feedback Revisions

DFM Considerations

- Designed parts with print orientation and support minimization in mind to reduce post-processing.
- Iteratively adjusted CAD models based on test fits and feedback to ensure optimal payload integration and CG alignment.

Collaborative Highlights

- Shared CAD models with team members for review and iterative improvement.
- Incorporated feedback from payload specialists and aerodynamicists to refine design.
- Documented each version to track design evolution and rationale for changes.

Engine Casing: Cradle-to-Grave Approach

- The rocket engine casing was initially hand-sketched and dimensioned on a whiteboard, leveraging rapid ideation and collaborative brainstorming.
- Critical measurements were taken and validated before CAD modeling, ensuring a single, successful 3D print iteration.
- The final engine casing was integrated into the rocket and performed successfully during flight, demonstrating effective DFM and cross-disciplinary teamwork.

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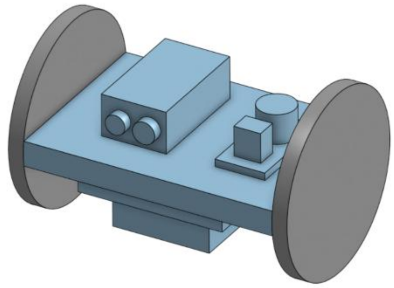
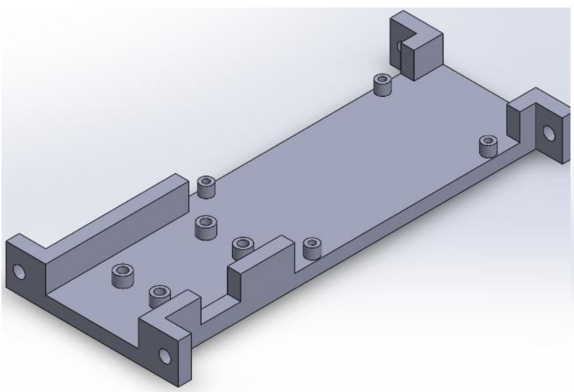
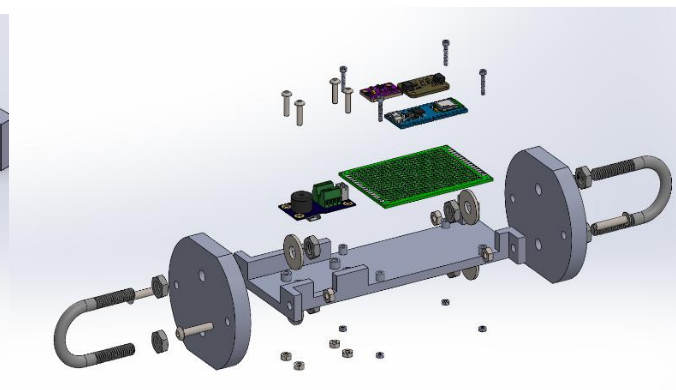
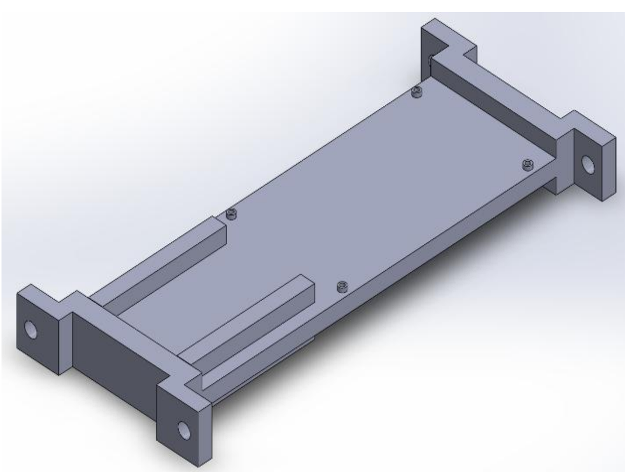


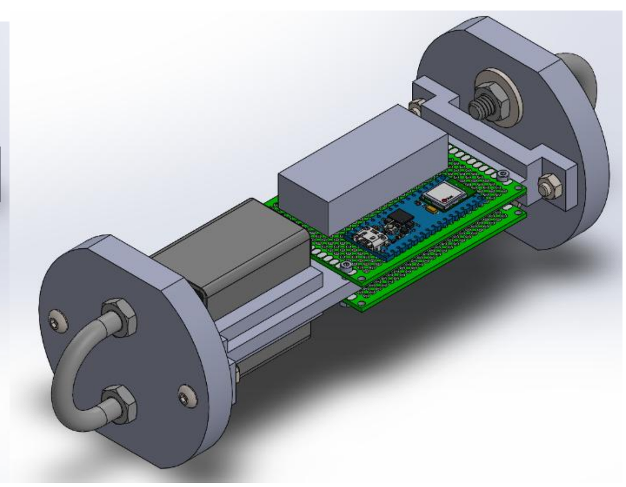
3D Printed Avionics Tray CAD Version 1





3D Printed Avionics Tray CAD Version 2





3D Printed Avionics Tray CAD Version 3

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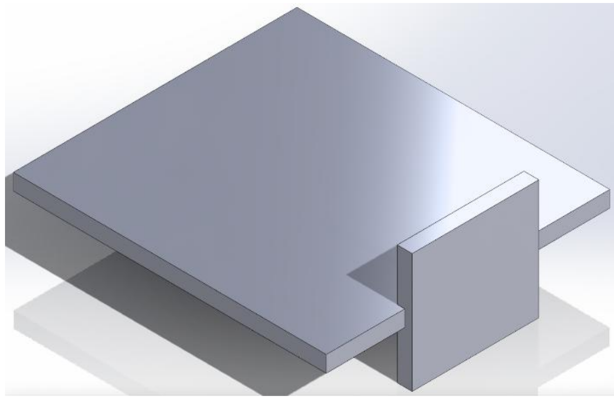
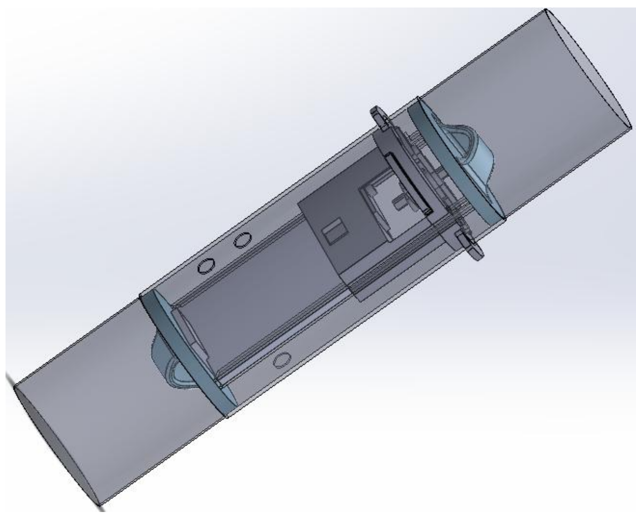


3D Printed Avionics Tray CAD Version 4



3D Printed Avionics Tray Packaged with Air Brake Assembly Final CAD

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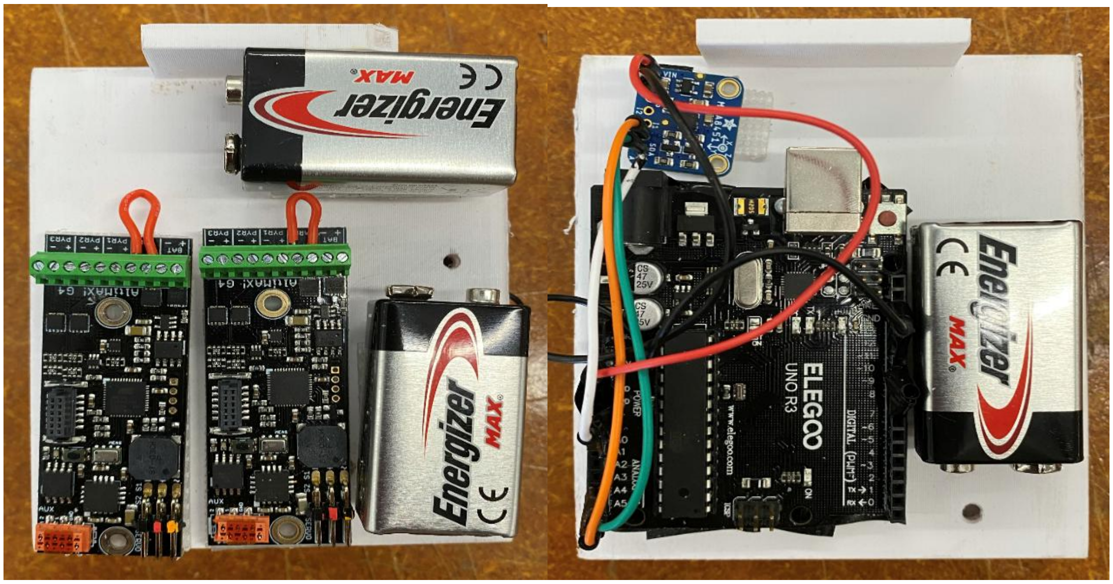
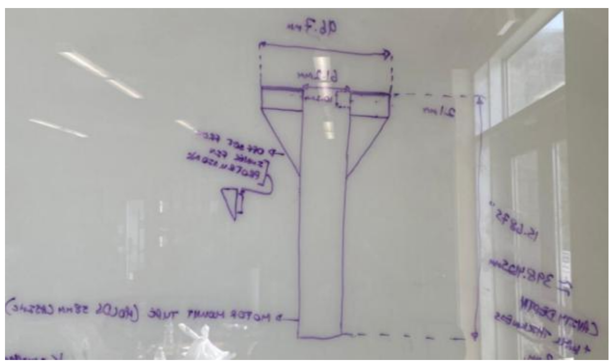
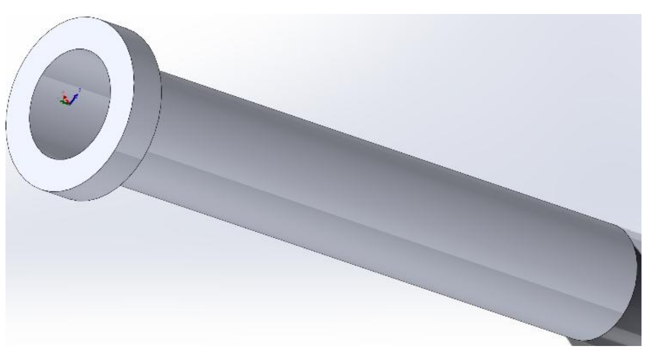


Assembled 3D Printed Avionics Tray



Hand Drawing to Dimension 3D Printed Engine Casing Before Creating CAD Model



3D Printed Engine Casing CAD

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Conclusion

These projects collectively demonstrate advanced knowledge of DFM principles across multiple manufacturing methods and materials. The portfolio highlights the importance of clear communication, iterative design, and collaborative problem-solving in achieving robust, manufacturable solutions for both industrial and high-performance applications.